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Research Experience

Research Internship | The University of Queensland, Australia **Sept 2026 - Present**
Objective: *To design a better architecture inspired by graph-based LLM reasoning methods under Professor Moloud Abdar*
- Reproduced **GFM-RAG** and **G-Reasoner**, **GNN-based** multi-hop retrievers for LLM reasoning, re-running training on a subset of data
- Exploring **self-distillation**, **BF16 training**, and **pre-training ablations** to boost retrieval accuracy and cut training/inference cost

Visiting Undergraduate Researcher | The Pennsylvania State University, USA **May 2026 - Jul 2026**
Objective: *To design and prototype an IFC-based security layer for multi-agent systems like OpenClaw under Professor Peng Liu*
- Co-authored a paper showing **classical Mandatory Access Control** is enforceable but impractical for multi-agent OpenClaw deployments
- Built three Discord-based **BLP** case studies; policy edges grew from **8 to 42** (2 to 3 agents), forcing manual routing and declassification
- Contributed to **HiStar++**, an in-process **IFC runtime** in OpenClaw enforcing label, taint, and clearance checks across various tool calls
- Owned the tool-to-object group (`runBeforeToolCall`) of four hook groups, enforcing **No-Write-Down** via block/approval verdicts
- Implementing a **declassification orchestrator** that finds and applies downgrade gates from L_{in} to L_{out} with human-approval fallback

Research Internship | Deakin University, Australia **May 2025 - Jul 2025**
Objective: *To safeguard privacy in agent-to-agent communication through research under Professor Ye Zhu*
- Built two **MCP servers** with **9 tools**: a local server for file access and doc search, and a remote server for contacts, calendar and mailing
- Deployed the remote server via SSH (Streamable HTTP) and ran the local one over stdio, connected to Cursor, Continue, and Claude Desktop
- Demonstrated **MCE**, **RAC** and **CT** via direct and **RADE** attacks on Claude and Llama-70B, documenting how injections fooled the models

Internships

Applied AI Engineering Intern | NephroAI, Remote **Mar 2026 - May 2026**
Objective: *To automate and accelerate DevOps triage by deploying a multi-agent AI system for bug analysis*
- Architected a **multi-agent DevOps triage** system using **CrewAI** and **FastAPI** to auto-resolve Azure DevOps (ADO) support tickets
- Engineered **GitHub MCP** tools, bypassing **30 req/min** rate limits via prompt-level **API batching** and **SSE timeout** handling
- Designed **vector retrieval pipelines** on **Azure AI Search** to auto-resolve recurring tickets from history, reducing **LLM API costs**
- Enforced **zero-hallucination** guardrails by isolating **Planner**, **Explorer**, and **Synthesizer** agent responsibilities across the pipeline

Projects

VeriRAG | Self-Verifying Document QA System | Self Project **Oct 2026**
Objective: *To build a document-QA system that verifies every generated claim against its sources and abstains instead of hallucinating*
- Built a **LangGraph** RAG pipeline with **hybrid dense + BM25** retrieval (top-20 each), RRF-fused and reranked to 8 chunks
- Designed a **two-gate verification critic** (coverage, groundedness) using a **DeBERTa NLI** model, with bounded regeneration and abstention
- Built **PyMuPDF** ingestion: tables to **Markdown** chunks, per-page text, and vision captions for embedded figures and vector charts
- Evaluated on **MMLongBench-Doc**, where the gated system declined **every unanswerable question** instead of fabricating any answer

Kaggle Competition | CommonLit - Evaluate Student Summaries **Dec 2025**
Objective: *To develop a hybrid NLP pipeline that scores student summaries on content accuracy and wording quality*
- Engineered **length, source-overlap, and readability** features from summaries and trained a **LightGBM** baseline at **0.577** MCRMSE
- Benchmarked **BERT**, **RoBERTa**, and **DeBERTa-v3** with prompt-grouped CV to simulate unseen test prompts; DeBERTa-v3 scored best
- Fine-tuned **DeBERTa-v3** on prompt-text-summary inputs; **last-4-layer mean pooling** over summary tokens cut MCRMSE 1.25 → 0.46
- Stacked **DeBERTa** out-of-fold predictions with features in **LightGBM** under **GroupKFold**, lowering MCRMSE from 0.536 to **0.508**

Competitions

Voice-Native Hospital Call Agent | DataForge **Sept 2026**
Objective: *To build a low-latency voice agent that handles live hospital calls end to end, with clean barge-in and graceful failure recovery*
- Built a **LangGraph** voice agent with **24 tools** over a **29-table SQLite** DB, logging per-turn latency, tool calls and end-of-call summaries
- Engineered turn-taking with **Silero VAD**, an adaptive noise gate and token-fenced **barge-in** that cancels stale LLM and tool-call output
- Designed failure handling so calls never go silent: on LLM failures, apologise, save context for same-day callback and end gracefully
- Deployed a demo on **AWS EC2** (**FastAPI WebSocket**, **Caddy** HTTPS), a browser-callable bridge designed to switch to **Twilio** later

General Championship Data Analytics | Technology Students Gymkhana | Gold **Feb 2026**
Objective: *To build an AI copilot that democratises data access by converting natural language into SQL and charts.*
- Built a **clarification-aware ReAct** agent over six **tool calls**, serialising state to **PostgreSQL** and batching SQL via **asyncio.gather**
- Built a **KPI DSL** of **JSON** rules with formula, significance, and parameterised **SQL**, eliminating metric hallucination in the agent loop
- Enforced **query-level tenant isolation** by wrapping every **LLM-generated** SQL in **CTE-scoped** aliases, blocking cross-tenant leakage

Amazon ML Challenge 2025 | All India ML Hackathon **Oct 2025**
Objective: *Predicting marketplace product prices from catalogue text and product images (multimodal regression, SMAPE metric)*
- Built a **TF-IDF** and regex baseline (extracting quantities, units), achieving **60.69%** SMAPE with **5-fold GroupKFold** validation
- Replaced BERT with **EmbeddingGemma** text embeddings to better capture numeric cues, abbreviations, and semi-structured specs
- Swapped CLIP for **SigLIP 2-Giant** image embeddings, as its sigmoid loss removed CLIP's batch-wise scale drift in regression
- Designed **dual-path cross-attention** fusion (text↔image) with a 3-layer FFN head, trained directly on **SMAPE**, achieving **48.67%**

Relevant Coursework

Advanced Calculus | Linear Algebra, Numerical And Complex Analysis | Theory Of Partial Differential Equations| Probability And Statistics | Programming And Data Structures * | Operations Research | Econometric Analysis * | Statistics For Economics | Embedded Machine Learning